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4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```

import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Prepare data for calculation
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')
df['Sales'] = df['Quantity'] * df['Price']

# Find the month with the highest total sales
best_month = df.groupby('Month')['Sales'].sum().idxmax()
highest_sales = df.groupby('Month')['Sales'].sum().max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")

```

The screenshot displays a user interface for running code and testing it. At the top, performance metrics are shown: Average time is 0.035 s (34.67 ms) and Maximum time is 0.063 s (63.00 ms). Test results indicate that 1 out of 1 shown test case(s) passed and 2 out of 2 hidden test case(s) passed. Below this, 'Test case 1' is detailed with a 63 ms execution time and a 'Debug' button. It compares 'Expected output' and 'Actual output' for the file 'sales_data.csv'. Both outputs show 'Best month: 2025-01' and 'Total sales: \$1210.00'. At the bottom, there are tabs for 'Terminal' and 'Test cases'.

Metric	Value
Average time	0.035 s (34.67 ms)
Maximum time	0.063 s (63.00 ms)

Test Case	Status	Time
Test case 1	Passed	63 ms

Output Type	Value
Expected output	Best month: 2025-01 Total sales: \$1210.00
Actual output	Best month: 2025-01 Total sales: \$1210.00